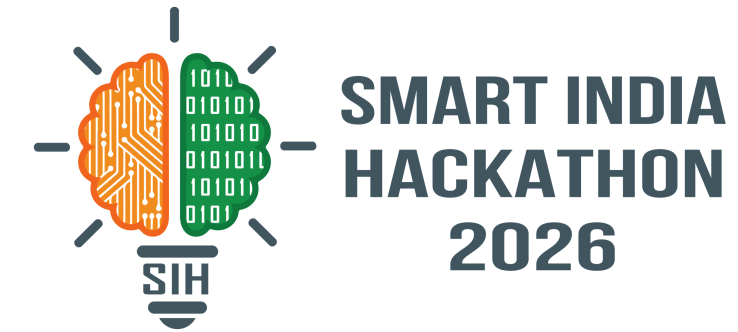


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- **Problem Statement ID –**
- **Problem Statement Title-**
- **Theme-**
- **PS Category- Software/Hardware**
- **Team ID-**
- **Team Name (Registered on portal)**



PROBLEM STATEMENT

The project aims to develop an AI-powered system that predicts fuel consumption and optimizes fleet operations using quantum-inspired algorithms to reduce fuel costs and greenhouse gas emissions while improving operational efficiency and sustainability.

- High fuel consumption and operating costs in fleet operations.
- Difficulty in accurately predicting fuel usage.
- Increased greenhouse gas emissions.
- Inefficient vessel allocation and route planning.
- Challenges in adopting alternative fuels.
- Need for intelligent, cost-effective, and sustainable fleet optimization.

KEY OBJECTIVES



Predict Fuel Consumption: Estimate fuel usage based on vessel type, load, speed, distance, route, and weather conditions.



Enable Green Fleet Operations: Evaluate LNG, methanol, hydrogen, ammonia, and shore power based on feasibility, cost, and lifecycle emissions.



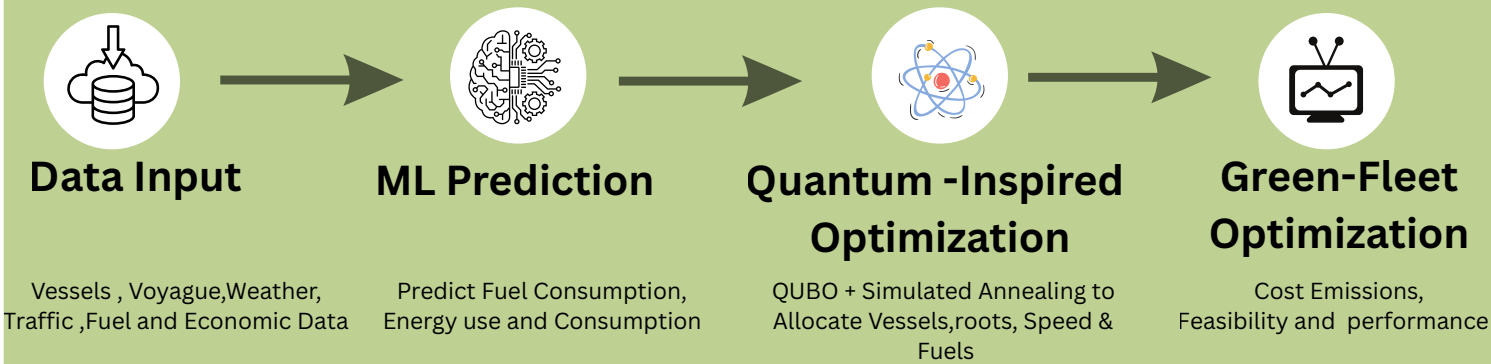
Minimize Costs and Emissions: Reduce fuel consumption, operating expenses, and greenhouse gas emissions.



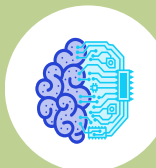
Support Decision-Making: Provide data-driven recommendations through an interactive dashboard.

PROPOSED SOLUTION

AI-powered platform combining machine learning with quantum-inspired optimization to predict fuel consumption and recommend efficient, cost-effective, and environmentally sustainable fleet operations.



KEY OBJECTIVES



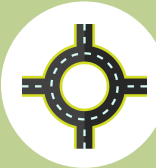
Fuel Consumption Prediction

Use machine learning algorithms to predict fuel consumption based on vessel speed, cargo load, weather, and route distance.



Quantum-Inspired Optimizer

Apply quantum-inspired algorithms to optimize vessel assignments, schedules, and fleet operations while satisfying operational constraints.



Route and Speed Optimization

Identify efficient routes and suitable cruising speeds to reduce fuel consumption, operating costs, and delays.



Green Technology Selection

Evaluate alternative fuels such as LNG, methanol, hydrogen, and ammonia, along with shore power, based on cost and lifecycle emissions.



Constraint Validation

Verify that optimized plans satisfy cargo capacity, delivery deadlines, vessel availability, and fuel compatibility requirements.

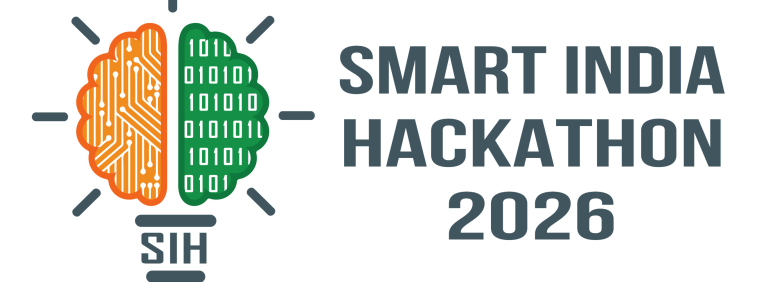


Interactive Dashboard

Display fuel predictions, operating costs, greenhouse gas emissions, and optimized fleet plans to support informed decision-making.

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TECHNICAL APPROACH



TECHNICAL APPROACH

Data Preparation
Collect and clean vessel,route,weather and operational data for analysis

Fuel Prediction (ML)
Use machine learning to Estimate fuel consumption , cost , time and emissions

Optimization (Quantum- Inspired)
Formulate fleet planning a QUBO problem and solve using simulated annealing

Route & Speed planning
Find efficient routers and speeds to minimize fuel use,cost and delays

Green Fuel Evaluation
Compare convential and alternative fuels (e.g,LNG , Methanol ,hydrogen ,ammonia, shore power) based on cost and emissions

Validation & Visualization
Check constraints and present results through an interactive dashboard

WORK FLOW

Raw Data
(Vessels ,routes,weather,fuel,etc)

Preprocessing
Clean + Transform + Preapare

ML Prediction
Fuel | Cost | Time | On-time

Quantum
QUBO + Simulated Annealing

Routes & Speed Planning
QUBO + Simulated Annealing

Green Fuel analysis
Cost | Emissions | Sustainability

Constraint
Capacity | Availability | Compatibilty

Dashboard
Compare | Decide | Optimize

TECHNOLOGY USED

Programming & Data handling

Machine learning

Quantum _inspired Optimization

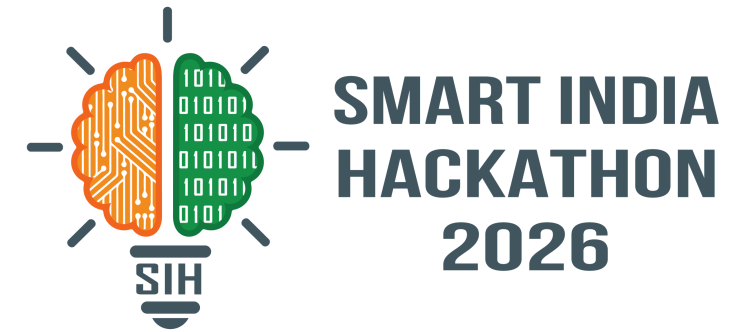
QUBO + Simulated Annealing

Visualization & Dashboard

Development Tools

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FEASIBILITY AND VIABILITY



Feasibility



Technology

Uses existing AI/ML and quantum-inspired optimization techniques.



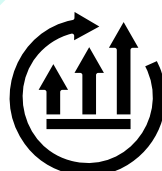
Data

Works with vessel, fuel, route, weather, and cargo data.



Development

Python, Scikit-learn, QUBO, Streamlit, and Plotly support rapid prototyping



Scalability

Can begin with a small fleet and expand to larger operations.



Validation

Results can be tested against traditional optimization methods.



Viability



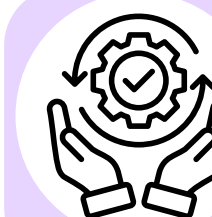
Cost Saving

Cost Saving Optimized routes, speeds, and vessel allocation can reduce fuel-related expenses.



Enviromental Benefits

Helps evaluate strategies for reducing greenhouse-gas emissions.



Operational Efficiency

Supports better scheduling, vessel utilization, and resource planning.



Industry Use

Applicable to shipping, logistics, and fleet-management operations



Future Scope

Can integrate real-time AIS, weather, fuel-price, and port data.

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IMPACT AND BENEFITS



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Scalable & Future-Ready -

The system can later integrate real-time vessel tracking, weather data, fuel prices, and larger fleet operations.

Fuel & Cost Reduction –
Optimized routes, speeds, and vessel allocation can reduce unnecessary fuel consumption and operating costs.

Lower Carbon Emissions –

Better fuel planning and evaluation of cleaner energy options can help reduce greenhouse-gas emissions

IMPACT BENEFITS

Sustainable Maritime Operations -

Supports comparison of alternative fuels and shore power based on cost, feasibility, and environmental impact

Better Decision Making –
AI-based predictions and an interactive dashboard provide data-driven recommendations for fleet operators.

Improved Fleet Efficiency-

Helps utilize vessels more effectively by matching vessel capacity with cargo and operational requirements

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RESEARCH AND REFERENCES



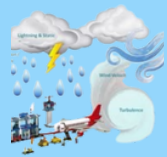
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RESEARCH



Fuel Prediction: ML models predict vessel fuel consumption using speed, engine, and operational data.



Weather Impact: Wind, waves, currents, and weather conditions affect fuel consumption and voyage efficiency.



Green Optimization: Fleet planning can minimize fuel cost and emissions while considering operational constraints.



Quantum Optimization: QUBO/Ising models can solve complex fleet allocation and scheduling problems



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